Product suggestions and customer groups: what we built

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In short. Our suggestions match what customers actually buy next about 1.5 times as often as showing our best sellers. The data the nightly job needs is 22 times smaller than the old table, and the whole job runs in under 2 seconds. For marketing, we found six customer groups, each with clear signature products.

Which method we chose, and why

We used a method called **truncated SVD**. It goes through two years of orders and finds 100 patterns in which products tend to be bought together. Every product and every customer is then described by 100 numbers instead of a row of 3,680 yes/no boxes. Products with similar numbers are bought by the same shops, which is exactly what a "customers who bought this also bought" box needs. We also tried the related method PCA on the same data. It gave almost the same results (40.2% against 40.9%) but needs an extra averaging step, so we kept SVD.

Does it work?

We held back the last two months of orders (9 Oct to 9 Dec 2011), built everything from the earlier orders only, and then checked our 10 suggestions for each of the 1,919 returning customers against what they really bought for the first time in those two months. For **41%** of them, at least one suggestion was something they went on to buy. Showing the 10 best sellers instead managed **27%**. On the White Hanging Heart T-Light Holder page, the box shows the red version of the same holder, the pink hanging heart candleholder, the hanging metal heart lantern, the cream heart card holder and the white metal lantern.

Size and speed

The compressed data is **7.3 MB**, against 161 MB for a full customer-by-product table. The whole nightly job (finding the patterns, building the box for all 3,680 products and picking 10 suggestions for each of 5,456 customers) took **1.5 seconds** on a laptop.

What we lose by compressing

The 100 patterns keep about 36% of the detail in the order history. Most of what gets dropped is one-off orders that no other shop repeats, and those don't help suggestions anyway. What we do lose is the ability to suggest products without an established history: only products bought by at least 10 customers are included.

Customer groups for marketing

Group	Shops	What they buy
Heart and wicker home decor	1,321	Hanging and wicker hearts, zinc T-light holders. Almost all UK (98%)
Traditional toys and crafts	945	Vintage paint sets, modelling clay, dominoes, harmonicas
Funny metal signs	935	Novelty metal signs. Narrowest range of products
Regency tea and baking	830	Regency teacups, teapots and cake tins
Kids' party tableware	718	Skull, dinosaur and spotty paper plates and cake cases. Most international group
Jumbo bags and storage	707	Jumbo bags in many prints. Widest range of products

The groups overlap: they describe what a shop leans towards, not hard boxes. A shop can fit more than one campaign.

What could go wrong

- **New products** are never suggested until at least 10 customers have bought them. That was about 1 in 10 of the new purchases in our test. Until then they should fall back to the best sellers in their category.
- **New customers** have no order history, so they can only be shown best sellers.
- **Season.** We tested on the two months before Christmas 2011 only. Rebuilding the patterns every night, as planned, keeps them current.
- **Popular items** get a head start in the suggestions, so we should check the box doesn't show the same few products everywhere.
- **Missing data.** 23% of order lines had no customer number and couldn't be used.
- **Past orders only.** This was tested on history. The real proof is a live test on the site.

Next steps

(1) Run an A/B test of the box on part of the product pages for four weeks. (2) Add the fallbacks for new products and new customers. (3) Marketing pilots one campaign per customer group. **Privacy:** no customer numbers appear in this memo or our slides. They stay in one internal file inside the team.